Clinical Study Summary Report (CSSR)
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1. Administrative & Study Identification

Full Study Title: Study of Pembrolizumab With Maintenance Olaparib or Maintenance Pemetrexed in First-line (1L) Metastatic Nonsquamous Non-Small-Cell Lung Cancer (NSCLC) (MK-7339-006, KEYLYNK-006) **Short Title/Acronym:** KEYLYNK-006 **Protocol Number:** MK-7339-006 **NCT Number:** NCT03976323 **Posting ID:** 681501866234303631 **Sponsor/Company Name:** Merck Sharp & Dohme LLC / Merck **Trial Status:** Active but not currently recruiting (ACTIVE_NOT_RECRUITING) **Investigational Product:** Pembrolizumab (Keytruda), Olaparib (Lynparza), Pemetrexed, Carboplatin / Cisplatin (Trade Name: Platinol, ATC Code: L01XA01) **Phase of Development:** Phase III **Medical Monitor:** Merck Global Clinical Development Lead

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2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate if pembrolizumab plus maintenance olaparib is superior to pembrolizumab plus maintenance pemetrexed with respect to Progression-Free Survival (PFS) per RECIST 1.1 by blinded independent clinical review (BICR) and Overall Survival (OS). * **Secondary Objectives:** To assess the number of participants experiencing adverse events (AEs), discontinuations due to AEs, and changes in health-related quality of life (EORTC QLQ-C30).

2.2 Background & Rationale To address the unmet medical need in metastatic nonsquamous NSCLC without targetable genetic alterations. Preclinical data suggested that PARP inhibitors (such as olaparib) upregulate PD-L1 expression, potentially increasing the efficacy of anti-PD-1 therapies like pembrolizumab. This study evaluates whether combining these modalities as a maintenance therapy following a first-line induction of chemo-immunotherapy can further improve survival outcomes.

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3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Active Comparator (Pembrolizumab + Pemetrexed) * **Blinding:** Open-label * **Randomization:** 1:1 Randomization (in the maintenance phase)

3.2 Planned Enrollment & Duration * **Target N:** 1003 participants * **Number of Sites:** Multinational * **Estimated Study Start:** 2019 * **Estimated Primary Completion:** January 2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically or cytologically confirmed diagnosis of Stage IV nonsquamous NSCLC. 3. Confirmation that epidermal growth factor receptor (EGFR), anaplastic lymphoma kinase (ALK), or proto-oncogene tyrosine-protein kinase (ROS1)-directed therapy is not indicated. 4. Measurable disease based on RECIST 1.1. 5. Have provided archival tumor tissue sample or newly obtained core or incisional biopsy of a tumor lesion not previously irradiated. 6. Have a life expectancy of at least 3 months. 7. Have a performance status of 0 or 1 on the Eastern Cooperative Oncology Group (ECOG) Performance Status assessed within 7 days prior to the administration of study intervention. 8. Have not received prior systemic treatment for their advanced/metastatic NSCLC. 9. Have adequate organ function. 10. Male and female participants who are not pregnant and of childbearing potential must follow contraceptive guidance during the treatment period and for 180 days afterwards. 11. Male participants must refrain from donating sperm, and female participants must refrain from donating eggs to others or freeze/store for her own use during the treatment period and for 180 days afterwards.

4.2 Exclusion Criteria 1. Has predominantly squamous cell histology NSCLC. 2. Has a known additional malignancy that is progressing or has progressed within the past 3 years requiring active treatment. 3. Has known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 4. Has a severe hypersensitivity ($\ge$ Grade 3) to pembrolizumab and/or any of its excipients. 5. Has a known hypersensitivity to any components or excipients of cisplatin, carboplatin, pemetrexed, or olaparib. 6. Has an active autoimmune disease that has required systemic treatment in past 2 years. 7. Has a diagnosis of immunodeficiency or is receiving chronic systemic steroid therapy. 8. Has a known history of human immunodeficiency virus (HIV) infection, a known history of hepatitis B infection, or known active hepatitis C virus infection. 9. Has interstitial lung disease, or history of pneumonitis requiring systemic steroids for treatment. 10. Has received prior therapy with olaparib or with any other polyADP ribose polymerization (PARP) inhibitor. 11. Has received prior therapy with an agent directed to PD-L1, anti PD-L2, or directed to a stimulatory or co-inhibitory T-cell receptor (e.g., CTLA-4, OX-40, CD137). 12. Has myelodysplastic syndrome (MDS)/acute myeloid leukemia (AML) or with features suggestive of MDS/AML. 13. Has history of a second malignancy, unless potentially curative treatment has been completed with no evidence of malignancy for 2 years. 14. Has completed palliative radiotherapy within 7 days of the first dose, with unrecovered toxicities or requirement for corticosteroids.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** * *Induction Phase (Up to 4 cycles):* Pembrolizumab 200 mg IV + Pemetrexed 500 mg/m^2 IV + Platinum (Carboplatin AUC 5 or Cisplatin 75 mg/m^2 IV) Every 3 Weeks (Q3W). * *Maintenance Phase:* Pembrolizumab 200 mg IV Q3W + Olaparib 300 mg Orally Twice Daily OR Pembrolizumab 200 mg IV Q3W + Pemetrexed 500 mg/m^2 IV Q3W. * **Route of Administration:** Intravenous (IV) and Oral. * **Duration of Treatment:** Up to 4 cycles of induction (~12 weeks) and up to 31 cycles of maintenance (total ~35 cycles or up to ~2 years). * **Second Course:** Qualified participants who complete up to ~35 cycles may be eligible to receive a second course of pembrolizumab for up to ~17 cycles (up to ~1 additional year).

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard supportive care for toxicities. * **Prohibited:** Prior treatment with a PARP inhibitor, concurrent investigational agents, or systemic immunosuppressive therapy.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to Day 0	Informed Consent, Biopsy/Archival tissue evaluation, Imaging
Baseline	Day 1	Enrollment, Start of Induction Therapy
Induction (C1-C4)	Q3W	IV infusions, Safety Labs, Response Evaluation at C4
Maintenance	Q3W	Randomization (for responders/stable disease), Dispense oral study drugs, ongoing efficacy/safety assessments
End of Study	Up to ~Year 2+	Final Efficacy Scans, Safety Check, Survival Follow-up

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Progression-Free Survival (PFS):** Defined as the time from randomization to the first documented progressive disease (PD) per RECIST 1.1 based on Blinded Independent Central Review (BICR) or death due to any cause. PD defined as $\ge$20% increase in the sum of diameters of target lesions, with an absolute increase of $\ge$5 mm, or appearance of new lesions. (Analyzed by Kaplan-Meier method in Maintenance Phase). * **Overall Survival (OS):** Defined as the time from randomization to death due to any cause. (Analyzed by Kaplan-Meier method in Maintenance Phase).

7.2 Secondary & Exploratory Endpoints * Number of Participants Who Experience One or More Adverse Events (AEs). * Number of Participants Who Discontinue Study Treatment Due to an AE. * Change From Baseline in European Organization for Research and Treatment of Cancer (EORTC) Quality of Life Questionnaire-Core 30

(QLQ-C30) Global Health Status / Quality of Life (Items 29 and 30) Scale Score.

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8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Standard collection evaluating immune-mediated AEs (e.g., pneumonitis, colitis) and chemotherapy/PARPi-related toxicities (e.g., anemia, thrombocytopenia). AEs during the second pembrolizumab course will not be counted towards safety outcome measures per protocol.
- **Serious Adverse Events (SAEs):** Reported per standard phase 3 oncologic trial timelines.
- **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee reviewing interim efficacy and safety data.

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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intention-to-Treat (ITT) for primary efficacy analyses; Safety Set for AE evaluations.
- **Sample Size Justification:** Sized to provide adequate power to detect statistically significant superiority in the dual primary endpoints of PFS and OS.
- **Handling of Missing Data:** Analyzed by the non-parametric K-M method. Participants without documented death at the time of analyses were censored at the date of last known to be alive.

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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
- **Monitoring Plan:** Standard site monitoring, continuous safety oversight, and independent radiologic review (BICR).
- **Audit Trail:** eCRF entry, site audits, and rigorous data clarification.

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11. Study Identifiers & Governance

- **Primary ID:** MK-7339-006
- **Registry Number:** NCT03976323
- **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC
- **Study Title:** KEYLYNK-006
- **Official Regulatory Title:** A Phase 3 Study of Pembrolizumab in Combination With Pemetrexed/Platinum (Carboplatin or Cisplatin) Followed by Pembrolizumab and Maintenance Olaparib vs Maintenance Pemetrexed in the First-Line Treatment of

Participants With Metastatic Nonsquamous Non-Small-Cell Lung Cancer (NSCLC)

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12. Clinical Framework

- **Disease Areas / Condition:** Carcinoma, Nonsquamous Non-small-cell Lung
- **Preferred Terminology / Semantic Type:** Nonsquamous non-small cell lung carcinoma / Carcinoma (Neoplastic Process)
- **Definition:** A malignant tumor arising from epithelial cells. Carcinomas that arise from glandular epithelium are called adenocarcinomas, those that arise from squamous epithelium are called squamous cell carcinomas, and those that arise from transitional epithelium are called transitional cell carcinomas (NCI Thesaurus).
- **Ontology Codes:**
- MeSH Code: D002277
- HPO Code: HP:0030731
- SNOMED ID: 1187425009 (Hierarchy: 1187225007 | Malignant epithelial neoplasm, 118285006 | Epithelial neoplasm, 108369006 | Neoplasm, 1240414004 | Malignant neoplasm)
- **Medical Methodology:** First-line chemo-immunotherapy induction followed by active maintenance.
- **Optimization Target:** Prolongation of PFS and OS.

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13. Study Procedures & Methodology

- **Management Pathway:** 4 cycles of induction followed by randomization into maintenance arms.
- **Education Protocol (HCP):** Standard protocol and safety monitoring training for trial investigators.
- **Education Protocol (Patient):** Trial overview, informed consent, and adverse event diary training.
- **Quality Audit Mechanism:** BICR for imaging, safety monitoring via DMC.

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14. Observation & Follow-Up Schedule

- **Total Duration:** Up to 35 treatment cycles (~2 years) plus potential second course up to ~17 cycles, and long-term survival follow-up.
- **Onsite Visit Cadence:** Every 3 Weeks (Q3W) correlating with treatment cycles.
- **Remote Monitoring Frequency:** Between treatment cycles or during long-term survival follow-up phases.
- **Checklist Review Interval:** Routine clinical assessments prior to each Q3W dosing cycle.

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]